PRO BONO SAR INITIATIVE // V1.6 MULTI-MODEL HARDENED ARCHITECTURE

Karhunkierros SAR & "Digital Wilderness Hut" (v1.6) — End-to-End Rescue Chain

Jam-Proof Aerial Lantern Rescue Drone (Holybro X500 V2 + Optical Diode + LiDAR/Optical Flow) & State Wilderness Hut Network Integration

Core Premise (Why This Saves Lives in Roadless Arctic Wilderness):

When a hiker or family becomes lost in roadless boreal or Arctic wilderness (such as Ruka / Oulanka's **Karhunkierros** trail in Finland), the nearest plowed road is often **8–25 km away**. **A police van or ambulance cannot drive into roadless forest**, and mobilizing a snowmobile or ski patrol (`Poliisi` / `Rajavartiolaitos` / `Pelastuslaitos` / `Vapepa`) takes **1.5–3 hours** — whereas in sub-zero temperatures (`-15 °C to -30 °C`), severe hypothermia sets in within **30–45 minutes**.

However, Finland's state forestry agency (**Metsähallitus Parks & Wildlife Finland**) maintains a network of **450+ open wilderness huts (*autiotupa*) and day huts** (equipped with a wood stove, dry firewood, and shelter) often just **0.5–2.5 km** away — **provided the lost party can find the right direction to the trail and hut in darkness, blowing snow, or GPS jamming.**



Figure 1: End-to-end wilderness rescue chain (Stages 1–5) — from the Fibox ARCA 48h passive trail counter and 112 SOS trigger to jam-proof search, 15 m warm cask drop, and above-canopy (28 m) Aerial Lantern guidance to a state wilderness hut.

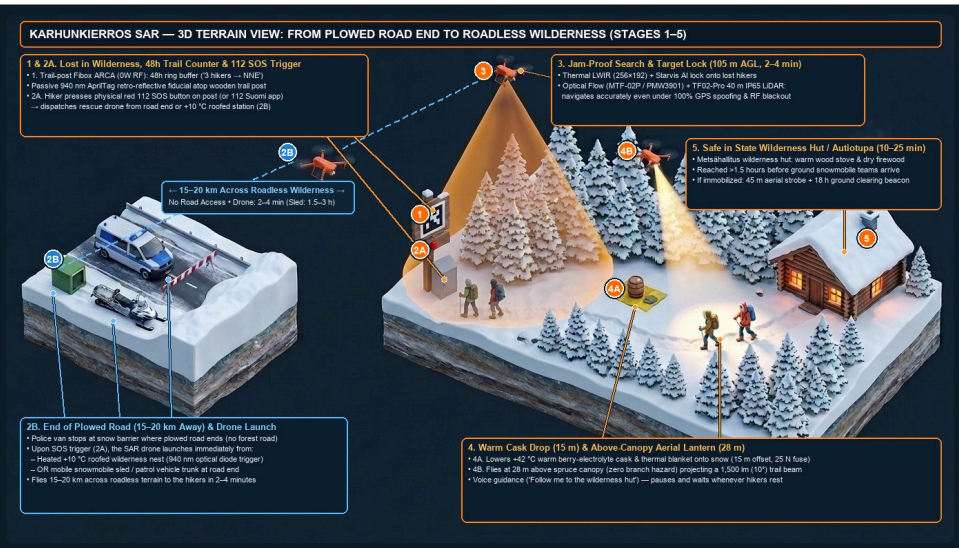


Figure 2: 3D isometric terrain and operational overview (Stages 1–5 in practice) across roadless wilderness unreachable by wheeled vehicles.

1. End-to-End Rescue Chain (Stages 1–5): From Lost on the Trail to a Warm Wilderness Hut

1. Passive Trail Counter	2. SOS Trigger & Launch	3. Jam-Proof Search (2–4 min)	4. Warm Cask (15 m) & Lantern (28 m)	5. Wilderness Hut (10–25 min)
Fibox ARCA 48h ring buffer (0W RF) + 940 nm AprilTag atop trail post.	112 / Trail-post SOS -> +10 °C nest (940 nm diode) OR mobile sled / van.	X500 at 105 m: Thermal/Starvis lock + Optical Flow / LiDAR without GPS.	+42 °C drink w/ 25 N fuse + 28 m above trees 1,500 lm (10°) beam.	Warm wood stove — or for injured: 45 m sky strobe + 18 h ground beacon.

Stage 1: Lost in Roadless Wilderness & Passive 48h Trail Counter (Fibox ARCA + 940 nm Fiducial)

- **Field Scenario:** A hiking group drifts off the trail in darkness or a snow squall 15–25 km from the nearest plowed road, where police vans cannot drive.
- **Background Preparation (`PassiveTrailSightingLedger`):** At the base of the wooden trail signpost sits an industrial weatherproof cabinet (`Fibox ARCA IP67`), a 2 m sensor pod (`PIR + 24 GHz radar + Starvis`), and a passive `940 nm` **AprilTag retro-reflective fiducial** atop the post. Operating 100% RF-silent (`0.00 W` RF), the unit records only an anonymous kinematic vector (*"3 hikers -> NNE"*) in a 48-hour volatile ring buffer that auto-purges after 48 hours unless an official SAR alert occurs.

Stage 2: Calling for Help, "Uber-for-SAR" Dispatch & Heated Roofed Nests (+10 °C)

- **Strictly Emergency-Triggered (`verify_humanitarian_sar_trigger`):** Activates only upon (1) a `112 Suomi` / `Garmin inReach` SOS ping, (2) pressing the trail post's **physical red** `112 SOS` **button** (works even when a phone battery has frozen dead at `-25 °C`!), or (3) an official search order (`SAR-ORDER`) from `Poliisi`, `Rajavartiolaitos`, or `Pelastuslaitos`.
- **Open Task Dispatch ("Uber-for-SAR") & Heated Roofed Stations (`SARDispatchMarketplaceCoordinator`):** The alert broadcasts as an open task to nearby **mobile drone units** (`Poliisi` vans at the plowed road end, `Rajavartiolaitos` snowmobile patrols, `Vapepa` units tapping *"Accept Task"*) and **heated roofed wilderness stations (`+10 °C` motorized shelter)**, which in `AUTO_ACCEPT_GEOFENCE` mode automatically accept local alerts (`<= 14 km`), open their motorized roof hatch, launch, and return under the roof to recharge. When multiple units accept, the system **auto-prioritizes** the fastest unit (#1 Primary, #2 Backup) while giving the human coordinator full dashboard override.

Stage 3: High-Speed Transit, GPS Jamming Bypass & Target Lock (2–4 Minutes)

- **Airframe (`Holybro X500 V2` carbon frame + `-40 °C 4S2P Li-Ion`):** High-visibility (**SAR Orange, RAL 2005**) quadcopter with quiet 10-inch propellers (`58 dBA`) flies at `105 m` AGL (`65 km/h`) to the target sector in **2–4 minutes**.
- **GNSS & RF Jamming Resilience (`ArduCopter EKF3`):** GPS jamming or spoofing cannot cause a flyaway: a downward **optical flow sensor (`MTF-02P / PMW3901`) and `Benewake TF02-Pro` (`40 m`, `IP65`) LiDAR** measure true ground velocity. If GNSS diverges from optical flow (`> 3.5 m/s`), the flight stack **immediately switches `EKF3` to optical-inertial navigation (`ekf_gnss_fenced_off_no_flyaway`)**. Under full RF blackout, the drone completes the rescue 100% autonomously onboard.

Stage 4: Warm Cask Drop (15 m) & Above-Canopy 28 m "Follow & Elastic Pull" Guidance

- **First-Aid Drop onto Open Snow (`15 m` Safety Offset + `25 N` Fuse):** The drone lowers onto open snow at least `15 meters` away an insulated warming cask and survival kit (`+42 °C` warm berry-glucose drink on tap, hand warmers, glucose gel packs, and thermal blanket) that releases via a `25 N` magnetic breakaway fuse, then climbs to `50 m` AGL before speaking instructions.
- **Above-Canopy `28 m` Elastic Pull-Guidance (`compute_elastic_pull_guidance_vector`):** To avoid spruce branches, `TF02-Pro` LiDAR terrain following keeps the drone safely **above the tree crowns (`28 m` AGL)**, tracking the hiker and "pulling" them along the trail by projecting a narrow `10°` (`1,500 lm`) **spotlight beam** `15 m` ahead toward the wilderness hut — sweeping closer if the hiker drifts off-trail (`> 8 m`) and pausing whenever the hiker rests.

Stage 5: Safe in State Wilderness Hut (10–25 min) — OR 18-Hour Beacon for an Injured Hiker

- **Ambulatory Party:** Guided by the spotlight beam, the hikers reach the state (**Metsähallitus**) wilderness hut and warm wood stove in **10–25 minutes** — over 90 minutes before a ground patrol arrives.
- **Injured Hiker Who Cannot Walk (`IMMOBILIZED_SHELTER_IN_PLACE`):** If the hiker cannot move within `3 minutes`, the drone announces: `"STAY IN PLACE INSIDE THE THERMAL BLANKET — GROUND RESCUE IS BEING GUIDED TO YOU"`, ascends to a `45 m` sky strobe (`1,200 lm`), and at `25%` battery lands in a clearing `15 m` away as an `18+ hour` **low-power ground beacon** until the snowmobile patrol arrives.

2. Timeline Comparison: Freezing Stationary vs. Aerial Lantern Guidance to a Wilderness Hut

Stage / Metric (12 km from road, -22 °C, dark forest + GPS jamming)	Without Rescue Drone (Waiting for Ground Patrol)	With Karhunkierros SAR (v1.6) Guide Drone (Guided to State Wilderness Hut)
1. Police Van / Ambulance Road Access	**Impossible** (`12.0 km` behind roadless ridge and river terrain)	**Not required for initial response** — launches from `+10 °C` station (`940 nm` optical diode) or snowmobile sled at road end.
2. First Contact & Operation Under GPS/RF Jamming	**`~135 minutes`** (`2 h 15 min` for snowmobile/ski patrol to arrive)	**`~2.5 minutes`** (operates without GPS or radio links via `MTF-02P / PMW3901` optical flow, `TF02-Pro` `40 m` LiDAR, and `940 nm` post fiducials).
3. First Aid & Thermal Energy Restoration	None available until ground patrol arrives	Lowers **onto snow** **`15 m`** **away** (`25 N` magnetic fuse) a **warm (`+42 °C`) berry-glucose drink cask, hand warmers, and thermal blanket**.
4. Guidance to Wilderness Hut (28 m Aerial Lantern)	Lost hikers cannot see the trail in darkness or tell which way the hut lies	Flies safely **above the trees (`28 m`)** **`15 m`** **ahead**, illuminates the trail with a **`1,500 lm`** **(`10°`) beam**, waits when hikers rest, and leads directly to the hut.
5. If Hiker Is Injured and Cannot Walk	High hypothermia risk in darkness without visual pinpointing	**`45 m`** **sky strobe beacon** + at `25%` battery lands in a clearing as an **`18+ hour`** **ground beacon** beside the sheltered casualty.
6. Time to Reach a Warm Stove Under a Roof	**`135 min`** *(exceeds the `30–45 min` hypothermia window several times over!)*	**`~10–25 minutes`** (**over `110 minutes` before ground teams arrive!**)

3. Privacy-by-Design Architecture ("Digital Wilderness Hut")

Even though the system exists purely to save human lives in emergencies, the architecture is engineered from the ground up so data-protection regulators, public right-of-access advocates (*Jokamiehenoikeudet*), and outdoorsmen can verify uncompromising privacy and safety:

1. **48-Hour Auto-Purging "Black Box" (`PassiveTrailSightingLedger`):** Trail-side units (`Fibox ARCA` , 100% shared hardware stack with `arctic-trail-cam`) are completely RF-silent (`0.00 W` RF), **never** use facial recognition or cloud uploads, and automatically overwrite anonymous movement vectors after 48 hours.
2. **Strictly Emergency-Triggered ("Pull, Don't Push"):** The drone launches **only** when a hiker triggers a `112 Suomi` / `Garmin inReach` SOS, presses the trail post's **physical red** **`112 SOS`** **button**, or when `Poliisi` , `Rajavartiolaitos` , or `Pelastuslaitos` issues an official search order (`SAR-ORDER`).
3. **Jamming-, Freeze-, and Icing-Resilient (`-25 °C` Polar Night, GPS Spoofing & Freezing Drizzle):** Uses `-40 °C` rated `8,000 mAh 4S2P` (`14.8 V` , `118.4 Wh`) Li-Ion cells, hydrophobic propeller coating + automatic ice-shedding spin bursts (`ICE_SHED_RPM_BURST`), a `940 nm` optical diode, and spoof-proof optical flow (`MTF-02P / PMW3901`) + `Benewake TF02-Pro` (`40 m` , `IP65`) LiDAR.
4. **Reindeer Herding Act (`Poronhoitolaki 848/1990 § 42`) Wildlife Protection:** Onboard AI distinguishes humans from reindeer and moose; when reindeer are present, the drone automatically enforces **`>= 100 m`** **altitude** and mutes its speaker.
5. **Hard-Coded** **`15 m`** **Safety Offset,** **`28 m`** **Canopy Floor &** **`~€1,189`** **Snap-In COTS Build (Zero Custom PCBs):** Software prevents flying closer than `15 m` to any person, keeps the drone at `28 m` above the trees, and assembles in **~2.5 hours without custom PCBs** from off-the-shelf COTS modules (`Holybro X500 V2` + `Pixhawk 6C` + `Pi 5` + `USB Thermal/Starvis` + `TF02-Pro 40m LiDAR` , total **`~€1,189`**).

4. Technical Architecture, COTS BOM (€1,189 Base / €1,329 v2.0) & Readiness Matrix

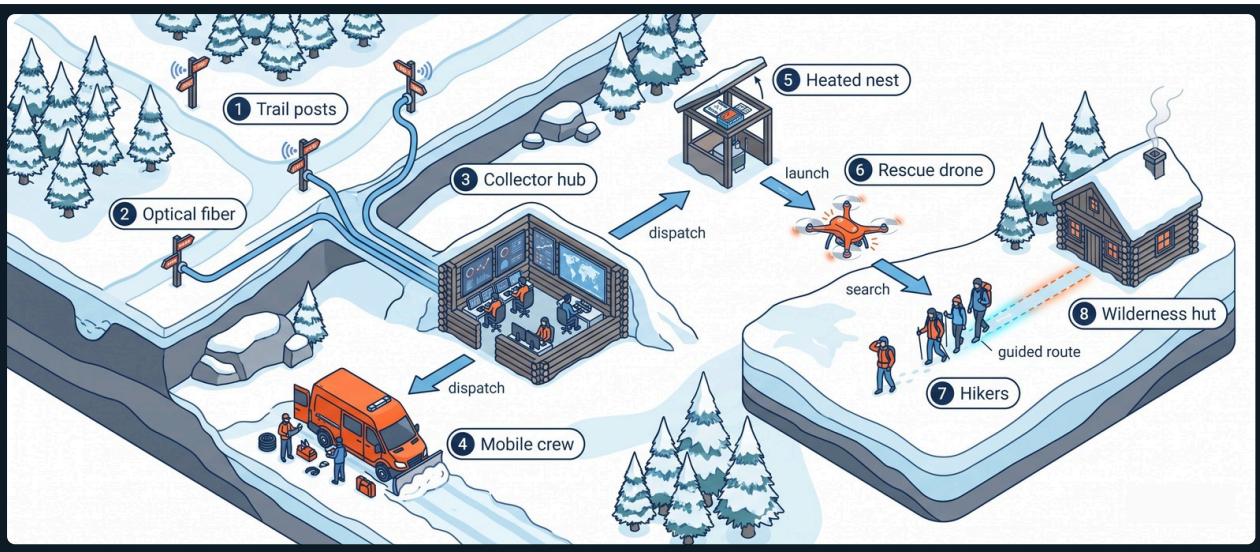


Figure 3: One system, eight parts working across the wilderness landscape — trail posts, optical fiber diode, collector hub, mobile crew, heated nest, rescue drone, hikers, and wilderness hut.



Figure 4: Snap-in off-the-shelf aircraft subsystems (~2.1 kg all-up weight, zero custom PCBs, ~2.5-hour bench assembly).

Subsystem Category	Off-the-Shelf Modules (JST-GH / STEMMA QT / USB-C / XT60)	Cost (EUR)
1. Airframe & Propulsion	`Holybro X500 V2` 500 mm carbon frame + `Pixhawk 6C` + `M10` GNSS + `4× 2216 KV920` motors + `1045` props + `RAL 2005` arms	**€490**
2. Navigation (<= 40 m AGL)	`Benewake TF02-Pro` (`40 m` , `IP65`) LiDAR + `MicoAir MTF-02P` (`PMW3901`) optical flow	**€134**
3. Companion Compute	`Raspberry Pi 5` (`8 GB`) + active cooler + `27W` buck (`5.1V/5A`) + `SanDisk Industrial 64 GB`	**€117**
4. Dual Vision (EO + LWIR)	`256×192 LWIR` USB-C thermal camera (`25 Hz`) + `Sony IMX462 Starvis` low-light USB camera (`130° FOV`)	**€235**
5. Telemetry & Override	`Holybro SiK / RFD868x` (`868 MHz`) or `900 MHz` LoRa + `RadioMaster ER5A V2` (`2.4 GHz ELRS`)	**€50**
6. Lantern, Audio & Drop	`1,500 lm` (`10°`) spotlight + `1,200 lm` strobe + `80 dB` speaker + `25 N` servo release + `+42 °C` care cask	**€145**
7. Arctic Battery (-40 °C)	`8,000 mAh 4S2P` (`14.8 V` , `118.4 Wh`) `-40 °C` Li-Ion (`XT60`) + `2.5 mm` silicone pad + `M3` kit	**€113**
Base Core Drone Subtotal	**Complete v1.6 snap-in COTS drone assembly (~2.5 hours, zero custom PCBs)**	**€1,189**
8. v2.0 Hardening (FIN-LAB)	Optional `LTE` burst modem (`+€95`) + `31-material` EMI ferrite kit (`+€15`) + wireless `UHF/DECT` drop-cask mic (`+€30`)	**+€140 (€1,329)**

5. Five-Layer Navigation, Four-Tier Link Ladder, Canopy Geometry & Verification Status

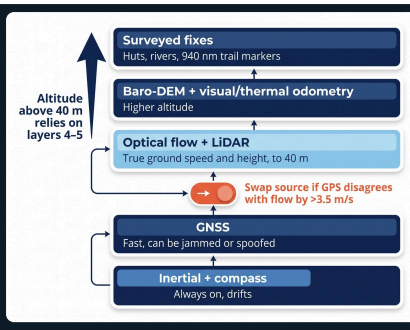


Figure 5: Five-layer navigation stack (optical flow + LiDAR to 40 m; Baro-DEM + VIO/TIO & 940 nm fixes above 40 m).

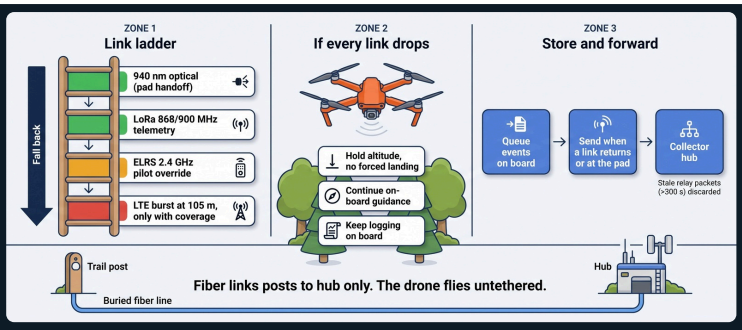


Figure 6: Four-tier link ladder (940 nm pad optical, 868/900 MHz LoRa, 2.4 GHz ELRS, LTE burst) & store-and-forward.

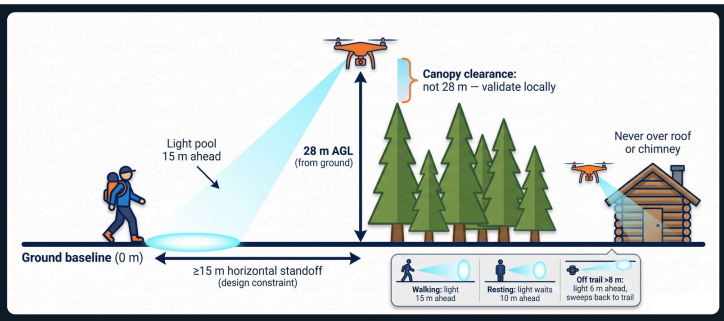


Figure 7: Above-canopy (28 m AGL) Aerial Lantern and >=15 m horizontal safety standoff geometry (walking, resting, off-trail >8 m, and cabin clearing).

Subsystem / Capability	Designed	Simulated / Unit-Tested (29/29)	Hardware Bench	Field-Validated	Authorised
Trail Posts & Collector Hub (`arctic-trail-cam`)	Yes	Yes	Yes (`Pi 5` + `Pico` + `LoRa`)	Partial (winter optical/RF)	No (`Metsähallitus` permit)
Core SAR Drone (`X500 V2` + `Pixhawk 6C` + `Pi 5`)	Yes	Yes (`37` `ArduCopter 4.5+` params)	Bench check (`drone_observer.py`)	No (`-25 °C` flight test open)	No (`Traficom SORA`)
GNSS Spoof Fence & <= 40 m Optical Flow / LiDAR	Yes	Yes (`F01-F53` regression suites)	Sensor bench check	No (live jamming test open)	No
Above-40 m Baro-DEM & VIO/TIO (`FIN-LAB-01`)	Yes	Modelled	Not yet flight-tested	No	No
Aerial Lantern (28 m AGL, >=15 m Standoff) & 25 N Drop	Yes	Yes (`compute_elastic_pull_guidance_vector`)	Mechanical bench	No (canopy survey needed)	No
"Uber-for-SAR" Dispatch & +10 °C Roofed Nest	Yes	Yes (`SARDispatchMarketplaceCoordinator`)	Dashboard & API verified	No (winter roof test open)	No (`Poliisi/RVL` MoU)

6. Pro Bono Pilot Opportunity & Phased Rollout (Phases 1–3)

1. Phases 1–2 (Private Forest Field Test — "Lean-To / Cabin = Wilderness Hut"): Validate first with mobile trunk/sled launches in dense coniferous forest (`€350` camera walk test -> `€1,189–€1,329` full COTS drone pilot under a safety pilot's RC override).
2. Phase 3 (Northern Finland Pro Bono Pilot): Provide the validated reference package for joint use with **Metsähallitus Parks & Wildlife Finland** (*Oulanka / Karhunkierros*, *UKK National Park*, *Pallas-Yllästunturi*), **Vapepa**, **Finnish Air Rescue Society (SLPS)**, **Poliisi**, **Rajavartiolaitos**, and regional **Rescue Departments**.